Low-Resolution Fully Polarimetric SAR and High-Resolution Single-Polarization SAR Image Fusion Network

Liupeng Lin, *Graduate Student Member, IEEE*, Jie Li, *Member, IEEE*, Huanfeng Shen, *Senior Member, IEEE*, Lingli Zhao, *Member, IEEE*, Qiangqiang Yuan, *Member, IEEE*, and Xinghua Li, *Senior Member, IEEE*

***Abstract*— The data fusion technology aims to aggregate the characteristics of different data and to obtain products with multiple data advantages. To solve the problem of reduced resolution of polarimetric synthetic aperture radar (PolSAR) images due to system limitations, we propose a fully PolSAR images and single-polarization synthetic aperture radar (SinSAR) images fusion network to generate high-resolution PolSAR (HR-PolSAR) images. To take advantage of the polarimetric information of the low-resolution PolSAR (LR-PolSAR) images and the spatial information of the high-resolution single-polarization SAR (HR-SinSAR) images, we propose a fusion framework for joint LR-PolSAR images and HR-SinSAR images and design a cross-attention mechanism to extract features from the joint input data. Besides, based on the physical imaging mechanism, we designed the PolSAR polarimetric loss functions for constrained network training. The experimental results confirm the superiority of the fusion network over traditional algorithms. The average peak signal-to-noise ratio (PSNR) is increased by more than 3.6 dB, and the average mean absolute error (MAE) is reduced to less than 0.07. Experiments on polarimetric decomposition and polarimetric signature show that it maintains polarimetric information well.**

***Index Terms*— Cross-attention mechanism (CroAM), fully polarimetric synthetic aperture radar (PolSAR), fusion, polarimetric loss, RCNN, single-polarization SAR (SinSAR).**

I. Introduction

THE polarimetric synthetic aperture radar (PolSAR) imaging system can obtain rich texture information and polarimetric information under day-and-night and all-weather conditions, so PolSAR images have been widely used in the field of geosciences, such as land use and land cover [1], [2], ship detection [3], [4], building damage assessment [5], [6], crop monitoring [7], [8], classification [9], [10], and so on. However, due to the complexity of the SAR imaging system and the limitations of hardware equipment, it is inevitable to the tradeoff between polarimetric information, image swath, and image resolution, when acquiring SAR images. For some SAR sensors, to obtain PolSAR images with multiple polarimetric information, the resolution of the images is reduced. This case means that the resolution of PolSAR images acquired by these sensors is often lower than that of single-polarization SAR (SinSAR) images. The reduction in resolution will further affect its application in practice. To solve the problem of the low resolution of PolSAR images, some scholars use frequency-domain methods to enhance spatial resolution. In 2001, Pastina *et al.* [11] first proposed to introduce polarimetric information into the super-resolution of PolSAR images. They used modern spectral analysis (SPECAN) techniques to perform single-channel super-resolution of PolSAR images to improve the spatial resolution. Suwa and Iwamoto [12] proposed a 2-D bandwidth extrapolation (2-D-PBWE) technique by extending the traditional bandwidth extrapolation method from SAR images to PolSAR images. By using the 2-D linear model, they extrapolated spatial frequency bandwidth from the azimuth direction and the distance direction to improve spatial resolution. These methods can reconstruct higher spatial resolution, but do not consider the relationship between images under different polarizations.

Another part of scholars uses the prior information or characteristics of the image itself to enhance the PolSAR image resolution. Jiong and Jian [13] used the projection onto convex sets (POCS) algorithm to construct a convex set to extract information from different polarimetric channels of low-resolution SAR images. The high-resolution polarimetric image for ship detection is generated by fusing different polarimetric channel information, but this method discards the polarimetric and phase information of the original SAR image during the processing [14]. Zou *et al.* [15] proposed a super-resolution method based on target decomposition and quadrant pixels (SRQP) to improve PolSAR images' spatial resolution. First, PolSAR images are decomposed into multiple

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Liupeng Lin is with the School of Resource and Environmental Sciences, Wuhan University, Wuhan 430079, China (e-mail: linliupeng@whu.edu.cn).

Jie Li is with the School of Geodesy and Geomatics, Wuhan University, Wuhan 430079, China (e-mail: aaronleecool@whu.edu.cn).

Huanfeng Shen is with the School of Resource and Environmental Sciences, Wuhan University, Wuhan 430079, China, and also with the Collaborative Innovation Center of Geospatial Technology, Wuhan University, Wuhan 430079, China (e-mail: shenhf@whu.edu.cn).

Lingli Zhao and Xinghua Li are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: zhaolingli@whu.edu.cn; lixinghua5540@whu.edu.cn).

Qiangqiang Yuan is with the School of Geodesy and Geomatics, Wuhan University, Wuhan 430079, China, and also with the Collaborative Innovation Center of Geospatial Technology, Wuhan University, Wuhan 430079, China (e-mail: qqyuan@sgg.whu.edu.cn).

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polarimetric components and then use the adjacent pixels of the center pixel of each polarization component to perform a weighted evaluation on the subpixels of the center pixel to obtain the super-resolution results. This method only considers the spatial correlation between a pixel and its neighbors, resulting in an obvious grid effect. To solve the grid effect problems of SRQP, the super-resolution method based on polarimetric spatial correlation (SRPSC) [14] is proposed. This method uses the polarimetric spatial correlation between pixels to determine the initial value of the subpixel and iteratively solves it to obtain a convergent high-resolution PolSAR (HR-PolSAR) image. Based on polarimetric coherent target decomposition, this method can maintain the polarimetric information well. However, the super-resolution reconstruction performed under different polarimetric decomposition has different results, which further affects the accuracy of the super-resolution results.

In recent years, deep learning [16]–[22] is widely used in image reconstruction and restoration tasks. To date, there are relatively few PolSAR super-resolution algorithms using deep learning. For the first time, a multichannel PolSAR images super-resolution (MSSR) method [23] is proposed to introduce deep learning into the PolSAR images super-resolution (PSSR) reconstruction, which is used for processing multichannel PolSAR data simultaneously. However, it does not fully consider the polarimetric information and numerical characteristics of PolSAR images. Subsequently, the PSSR method [24] is proposed to solve this problem. They used a complex block, transposed convolution, parametric rectified linear unit (PReLU), and other structures to maintain polarimetric and numerical characteristics. This method can effectively enhance PolSAR image resolution and maintain polarimetric information.

In this article, inspired by fusion technology [25], [26], we propose a novel PolSAR images and SinSAR images fusion network (PSFN) to efficiently obtain HR-PolSAR images. We adopted a novel data fusion framework and designed the corresponding network structure according to the characteristics of the data. Besides, we designed a polarimetric loss function based on the physical imaging mechanism of the PolSAR data. The contributions of PSFN are as follows.

1) Fusion framework for low-resolution PolSAR (LR-PolSAR) images and high-resolution single-polarization SAR (HR-SinSAR) images. Different from the existing PolSAR images resolution reconstruction methods, this strategy can fully integrate spatial texture information of the SinSAR images while maintaining polarimetric information of the PolSAR images.
2) Novel PolSAR attention mechanism. We designed the cross-attention mechanism (CroAM) to take advantage of polarimetric information of LR-PolSAR images and spatial information of HR-SinSAR images.
3) Loss functions based on the PolSAR physical imaging mechanism. The polarimetric loss function is used to intuitively constrain network training and enhance network interpretability.

The rest of this article is organized as follows. Section II introduces the PolSAR data and the detailed architecture of PSFN. In Section III, the experimental results on four datasets are presented. The polarimetric analysis experiments are provided in Section IV. The conclusions and prospects for future research are summarized in Section V.

II. Details of the Proposed Fusion Network

A. PolSAR Data Organization

The PolSAR imaging system can obtain the full polarimetric characteristics of the scatterers. For monostatic PolSAR data, it can be expressed in the form of 2×2 complex backscattering matrix [27], as shown in the following equation:

$$S = \begin{bmatrix} S_{\mathrm{HH}} & S_{\mathrm{HV}} \\ S_{\mathrm{VH}} & S_{\mathrm{VV}} \end{bmatrix} \tag{1}$$

where S_{HH} and S_{VV} represent power return in the co-polarized channels; S_{HV} and S_{VH} represent power return in cross-polarized channels. SinSAR data refer to SAR data that contain only one of S_{HH}, S_{VV}, S_{HV}, and S_{VH}, while fully PolSAR data refer to SAR data that measure the full scattering matrix. In the case of satisfying the reciprocity theory and ignoring the influence of system noise, the backscattering matrix can be represented by the Lexicographic feature vector

$$\Omega = \begin{bmatrix} S_{\mathrm{HH}} & \sqrt{2}S_{\mathrm{HV}} & S_{\mathrm{VV}} \end{bmatrix}^{\mathrm{T}} \tag{2}$$

where $^{\mathrm{T}}$ represents the transpose operation. The covariance matrix C_3 [27] is obtained by the outer product operation of the Lexicographic feature vector and its conjugate transpose vector

$$C_3 = \left\langle \Omega \cdot \Omega^{*\mathrm{T}} \right\rangle = \begin{bmatrix} \left\langle |S_{\mathrm{HH}}|^2 \right\rangle & \sqrt{2}\left\langle S_{\mathrm{HH}} S_{\mathrm{HV}}^* \right\rangle & \left\langle S_{\mathrm{HH}} S_{\mathrm{VV}}^* \right\rangle \\ \sqrt{2}\left\langle S_{\mathrm{HV}} S_{\mathrm{HH}}^* \right\rangle & 2\left\langle |S_{\mathrm{HV}}|^2 \right\rangle & \sqrt{2}\left\langle S_{\mathrm{HV}} S_{\mathrm{VV}}^* \right\rangle \\ \left\langle S_{\mathrm{VV}} S_{\mathrm{HH}}^* \right\rangle & \sqrt{2}\left\langle S_{\mathrm{VV}} S_{\mathrm{HV}}^* \right\rangle & \left\langle |S_{\mathrm{VV}}|^2 \right\rangle \end{bmatrix} \tag{3}$$

where $\langle \cdot \rangle$ represents the spatial statistical averaging operator. $*$ represents the conjugate operation. C_3 can be represented by the elements of the upper triangular matrix and can be represented as a 1×9 real-valued vector

$$C_{\mathrm{value}} = [\mathrm{Re}_{11}, \mathrm{Re}_{12}, \mathrm{Im}_{12}, \mathrm{Re}_{13}, \mathrm{Im}_{13}, \mathrm{Re}_{22}, \mathrm{Re}_{23}, \mathrm{Im}_{23}, \mathrm{Re}_{33}]^{\mathrm{T}} \tag{4}$$

where Re represents the value of the real part, Im represents the value of the imaginary part, and the subscript represents its position in the C_3 matrix.

Due to system limitations and other degradation factors, the HR-PolSAR image $C_x \in \mathbb{R}^{\mathrm{HEI} \times \mathrm{WID} \times \mathrm{CHA}}$ is degraded into an LR-PolSAR image $C_y \in \mathbb{R}^{\mathrm{hei} \times \mathrm{wid} \times \mathrm{cha}}$ through the degradation model $C_y = f_{\mathrm{d}}(C_x)$, where $f_{\mathrm{d}}(\cdot)$ represents the downsampling operator. HEI, WID, CHA and hei, wid, cha are height, width, and channel numbers of HR-PolSAR image and LR-PolSAR image, respectively. $r_{\mathrm{azimuth}} = \mathrm{HEI/hei}$ and $r_{\mathrm{range}} = \mathrm{WID/wid}$ are the spatial resolution ratios of LR-PolSAR image and HR-PolSAR image in the azimuth and distance direction, respectively. The HR-SinSAR intensity image $I_x \in \mathbb{R}^{\mathrm{HEI} \times \mathrm{WID} \times 1}$ can be obtained by performing intensity extraction operator $I_{x,i} = \Im_i(C_x), i \in (\mathrm{HH}, \mathrm{HV}, \mathrm{VH}, \mathrm{VV})$ on HR-PolSAR image. $\Im(\cdot)$ represents the intensity image extraction operator and i represents different polarization channels.

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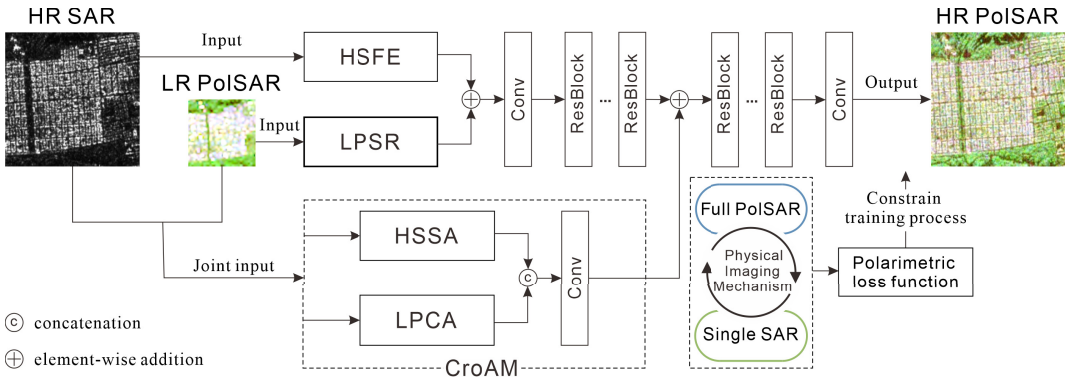

Fig. 1. Flowchart of the proposed method.

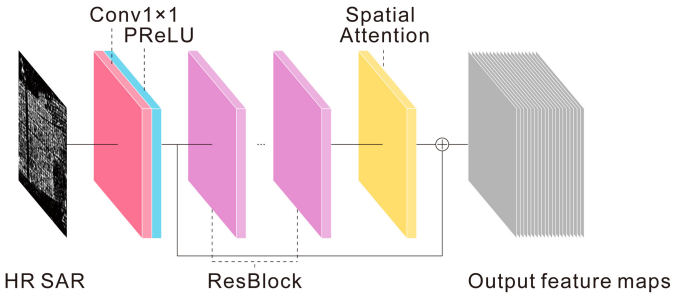

Fig. 2. HSFE module.

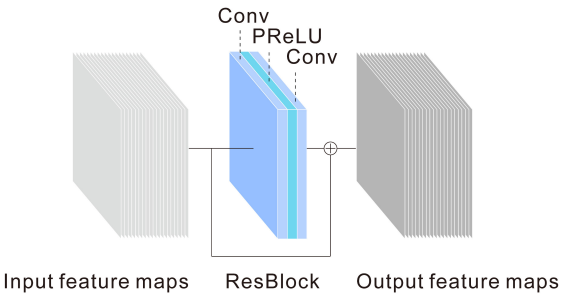

Fig. 3. Residual unit.

B. Fusion Framework for PolSAR and SinSAR

To overcome the reduction of PolSAR images resolution caused by the degradation process, we propose a novel PSFN. As illustrated in Fig. 1, the proposed fusion network is the end-to-end residual convolutional neural network. The input of the proposed method is the joint input of LR-PolSAR data and HR-SinSAR auxiliary data. The spatial features of HR-SinSAR images and polarimetric features of LR-PolSAR images are extracted by the HR-SinSAR feature extraction (HSFE) module and LR-PolSAR super-resolution (LPSR) module, respectively. Besides, a CroAM is designed to weigh the input data mutually and guide the extraction of spatial information and polarization information, which includes two submodules: HR-SinSAR spatial attention (HSSA) module and LR-PolSAR channel attention (LPCA) module. Among them, the HSFE module and LPSR module separately extract information from HR-SinSAR and LR-PolSAR, while the CroAM module uses HR-SinSAR and LR-PolSAR to extract information under mutual guidance. Through the separate and cooperative information extraction modules, while ensuring the effective information extraction of a single data, the guided spatial information and polarimetric information are extracted. In terms of the loss function, according to the physical imaging mechanism, a corresponding polarimetric loss function is designed to constrain the training process.

For this fusion framework, we use LR-PolSAR image C_y and HR-SinSAR intensity image I_x as the input to the network, and generate HR-PolSAR image C_x through the proposed network. HR-SinSAR images and LR-PolSAR images are used as the only input for feature extraction of the HSFE module and LPSR module, respectively. In the HSSA module and the LPCA module, the HR-SinSAR images and the LR-PolSAR images are simultaneously used as joint input data.

In the proposed fusion framework, the actual polarimetric information of the PolSAR images is used to correct the feature maps of the SinSAR images, and the high-precision spatial details of the SinSAR images are used to enhance spatial information of PolSAR images. The implementation details are provided below.

C. HSFE Module

As displayed in Fig. 2, in the HR-SinSAR image feature extraction module, first, one convolution layer is used to increase the channels of the SinSAR intensity images to fully extract spatial information of the SinSAR images. In this module, five residual units are used to extract features. The residual structure [28] was proposed to solve the problem of performance degradation in the deep network and accelerate network convergence [29]. In this article, the residual unit consists of three parts: 2-D convolution, PReLU activation function, and residual structure. The residual unit uses the 2-D convolutional layers to extract features, the PReLU activation function to enhance nonlinear characteristics, and the residual structure to connect the convolutional layers. The residual unit structure is illustrated in Fig. 3. The attention mechanism is widely used in computer vision tasks to recalibrate feature maps [30]–[34]. The spatial attention mechanism can recalibrate the high-frequency parts of the image with rich texture information and continuously increases the weight of high-frequency information through network training to achieve the goal of enhancing high-frequency information. In the proposed method, a spatial attention module is introduced into the HSFE module, which aims to better extract the spatial detail

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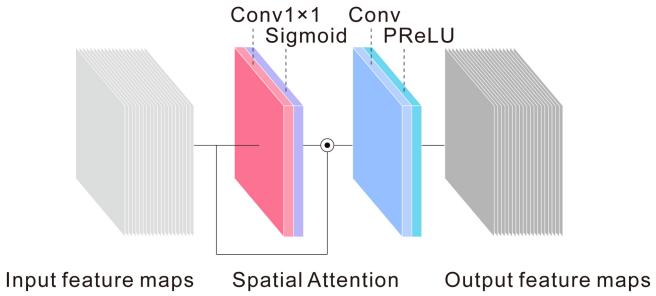

Fig. 4. Spatial attention module.

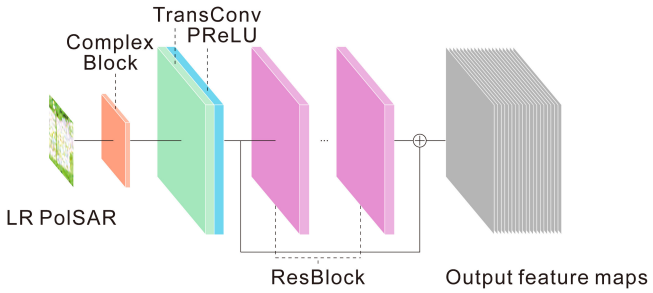

Fig. 5. LPSR module.

information of the SinSAR image. As shown in Fig. 4, in the spatial attention module, one convolution layer is used to distill the information in the channel direction of the feature maps extracted by the residual unit, and convert it into a 2-D feature map. Subsequently, the sigmoid function is adopted to normalize the feature map $\in \mathbb{R}^{\mathrm{HEI} \times \mathrm{WID} \times 1}$ to obtain a spatial weight layer. Finally, the spatial weight layer is multiplied by the residual unit feature maps as the weighted feature maps of the spatial attention mechanism. The spatial attention module can be expressed as follows:

$$F_{\mathrm{SA}} = F_{\mathrm{input}} \odot \sigma\left(W_{\mathrm{SA}} \circ F_{\mathrm{input}} + b_{\mathrm{SA}}\right) \tag{5}$$

where F_{SA} is output feature maps and F_{input} is input feature maps. W_{SA} is the convolution kernel of the spatial attention module, b_{SA} represents the bias of it, $\circ$ is convolution operation, $\sigma(\cdot)$ represents the sigmoid activation function, and $\odot$ is the element-wise multiplication operator. For the feature maps extracted by the HSFE module, we directly add them as auxiliary information to the LPSR module to enhance the spatial detail information of the LPSR module.

D. LPSR Module

In the LR-PolSAR image super-resolution reconstruction module, we make a series of improvements to the PSSR network. As shown in Fig. 5, deconvolution is used to up-sample the LR-PolSAR images to improve resolution and keep the complex blocks [24] in the PSSR network which are used to extract the PolSAR images mixed characteristics. Also, several residual units with the same configuration as in the HSFE module are used to replace the 20 convolutional layers in the PSSR network to speed up the training process of the super-resolution module.

In the PSSR network, deconvolution [35] is employed to up-sample the LR-PolSAR image first, and then complex blocks are used to extract mixed features from the up-sampled

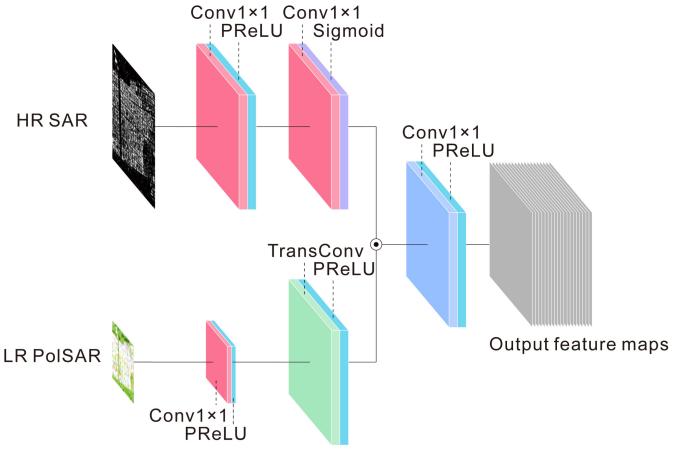

Fig. 6. HSSA module.

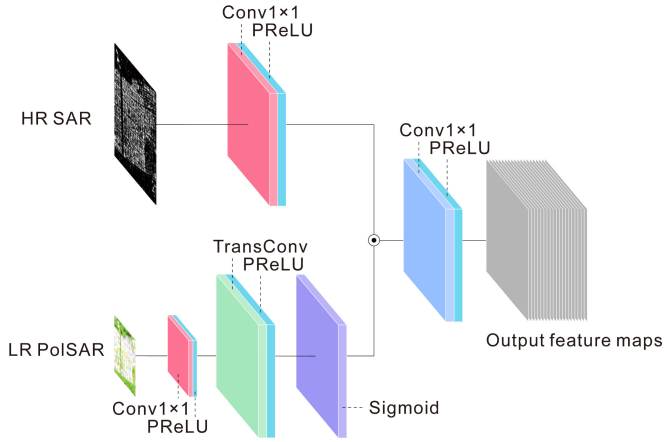

Fig. 7. LPCA module.

PolSAR images. In the case of insufficient convergence in the initial training stage, the process that first uses transposed convolution to up-sample the image and then employs the complex block to extract features may cause a certain error in the mixed features extracted from the complex block. Therefore, in the proposed method, we use complex blocks to extract mixed features from LR-PolSAR images and then use transposed convolution to up-sample the extracted feature maps. In this way, the accuracy of the feature maps obtained at the initial training stage is improved.

It should be noted that different from HSFE module, this fusion network does not use any attention module in the LPSR module. The essence of attention module is to assign a higher weight to the information of interest in the image, which will lead to the phenomenon that the information in some areas of the image is weakened. Our goal is to enhance the spatial texture information of LR-PolSAR images while maintaining its original polarimetric information. Therefore, we did not introduce the attention mechanism in the LPSR module, but used the output of the LPSR module as the basic information.

E. Cross-Attention Mechanism

To take advantage of the complementary information of LR-PolSAR images and HR-SinSAR images, a CroAM is designed to use respective advantages of LR-PolSAR data and HR-SinSAR data to enhance the polarimetric information and

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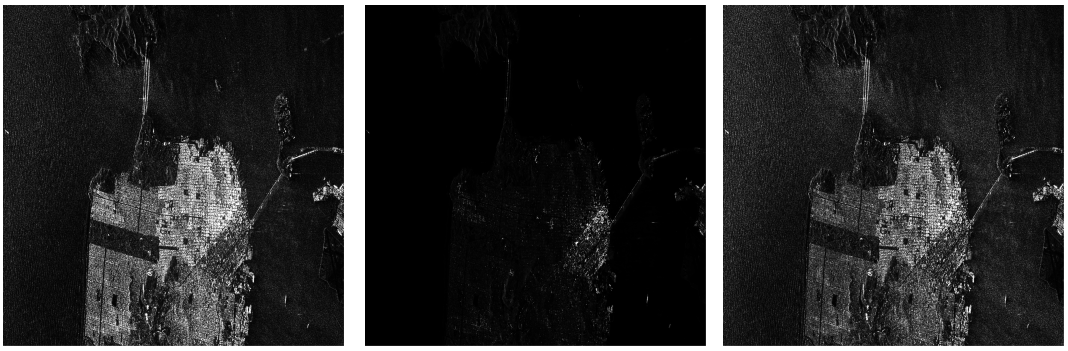

Fig. 8. Intensity images in different polarimetric modes.

spatial texture information of feature maps generated by the LPSR module. The mechanism consists of two parts: an HR-SinSAR spatial attention module and a low-resolution fully PolSAR channel attention module.

1) HSSA Module: Compared with LR-PolSAR images, HR-SinSAR images have higher accuracy in spatial information. Using its high-precision spatial information to guide and weight the feature maps extracted from LR-PolSAR images can make the spatial features extracted by LR-PolSAR images more accurate and reliable. The HSSA module aims to extract the spatial information weights of HR-SinSAR images and to use the weights to recalibrate the feature maps of LR-PolSAR images. The spatial information of HR-SinSAR images is introduced to enhance LR-PolSAR images. The structure diagram of the HSSA module is displayed in Fig. 6.

In the HSSA module, one convolutional layer is employed to reduce the dimensions of feature maps of HR-SinSAR images along the channel direction and then normalize them through the sigmoid function to generate a spatial weight layer. In contrast, we use transposed convolution to up-sample the LR-PolSAR images to $\in \mathbb{R}^{\mathrm{HEI}\times\mathrm{WID}\times\mathrm{CHA}}$. Subsequently, we multiply the spatial weight layer with the feature maps of the up-sampled PolSAR images to obtain the feature maps of the spatially weighted PolSAR images. The output of HSSA can be written as follows:

$$F_{\mathrm{HSSA}} = (W_{\mathrm{HSSA1}} \circ F_{\mathrm{LP}} + b_{\mathrm{HSSA1}}) \odot \sigma\,(W_{\mathrm{HSSA2}} \circ F_{\mathrm{HS}} + b_{\mathrm{HSSA2}}) \quad (6)$$

where F_{HSSA} is output feature maps of HSSA module; F_{LP} and F_{HS}, respectively, represent the feature maps generated by LR-PolSAR and HR-SinSAR. W_{HSSA1} and W_{HSSA2} represent the convolution kernel of the HSSA module; b_{HSSA1} and b_{HSSA2} represent the bias of the HSSA module.

2) LPCA Module: Compared with HR-SinSAR images, LR-PolSAR images have richer polarization information. If the PolSAR image is directly generated from a SinSAR image, it is obvious that the generated PolSAR image has lower reliability in terms of polarimetric information. Therefore, we use the actual polarimetric information of LR-PolSAR images to guide the generation of feature maps of SinSAR images, so that the feature maps extracted by HR-SinSAR have both high-precision polarimetric information and spatial texture information. The LPCA module is designed to extract the polarimetric information weights of LR-PolSAR images and guide the feature maps of HR-SinSAR. The polarimetric information of PolSAR is introduced to enhance the HR-SinSAR images.

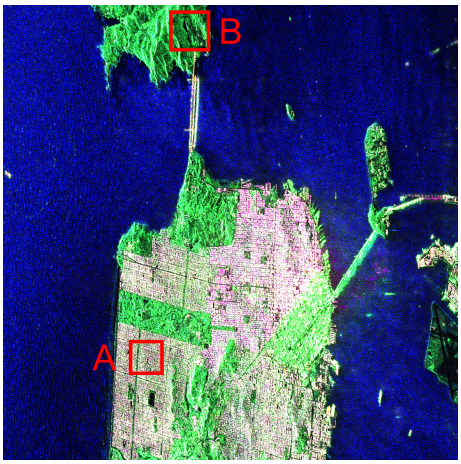

Fig. 9. PSFN fusion result for the RS-2 imagery (San Francisco).

As shown in Fig. 7, in the LPCA module, we use transposed convolution to up-sample the LR-PolSAR images and extract feature maps, and then use the channel attention mechanism to generate the channel weight layers. Subsequently, it is multiplied by the feature maps extracted from HR-SinSAR images to obtain feature maps of channel-weighted SinSAR images. Through the channel attention module, LR-PolSAR images are used to enhance the polarimetric information of SinSAR images.

The LPCA module can be written as follows:

$$F_{\mathrm{LPCA}} = (W_{\mathrm{LPCA1}} \circ F_{\mathrm{HS}} + b_{\mathrm{LPCA1}}) \odot \sigma\,(W_{\mathrm{LPCA2}} \circ F_{\mathrm{LP}} + b_{\mathrm{LPCA2}}) \quad (7)$$

where F_{LPCA} is output feature maps of the LPCA module. F_{LP} and F_{HS} are the feature maps generated by HR-SinSAR and LR-PolSAR, respectively. W_{LPCA1} and W_{LPCA2} represent the convolution kernel of the LPCA module; b_{LPCA1} and b_{LPCA2} represent the bias of the LPCA module.

Different from the existing channel attention mechanism, the channel attention mechanism in LPCA takes the large data dynamic range of PolSAR data and the complex data distribution into account. We remove the pooling operation that is widely used in existing channel attention modules to avoid numerical anomalies caused by mean pooling or maximum

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pooling. In the proposed channel attention module, we directly normalize the feature maps extracted by convolutional layer with the sigmoid function to obtain the channel attention weights. Subsequently, we multiply the weights with feature maps to get feature maps weighted by the channel attention mechanism. Different from the global channel attention weight of the existing channel attention modules, the proposed method uses local weights, which can better adapt to the characteristics of the large dynamic range of PolSAR data.

3) Information Fusion Strategy of HSSA and LPCA: After processing by the HSSA module and the LPCA module, spatially weighted feature maps and channel-weighted feature maps are obtained. To treat feature maps extracted by two modules equally, we use the cascading strategy for information fusion of the feature maps obtained by the two modules. The information fusion strategy can be written as follows:

$$F_{\text{fusion}} = \text{Concat}(F_{\text{HSSA}}, F_{\text{LPCA}}) \tag{8}$$

where Concat$(\cdot)$ is the cascade function, used to concatenate the feature maps generated by the HSSA module and LPCA module, respectively.

F. Fusion Loss Functions

In terms of loss functions, we design the loss functions from two aspects: the numerical value of the PolSAR images and the physical imaging mechanism of the PolSAR images. In terms of numerical value, mean square error (MSE) loss is adopted to constrain numerical values of the PolSAR images. For the training set $\{C_x^i, C_y^i, I_x^i\}_{i=1}^N$, we use transposed convolution to get the up-sampling results C_u^i of C_y^i, and use MSE loss for calculating loss between the residual layers $\Re^i = C_x^i - C_u^i$ and results of PSFN $f_{\text{net}}(C_y^i, I_x^i)$. Among them, N represents the image pairs number. C_x^i represents the HR-PolSAR image patches, C_y^i represents the LR-PolSAR image patches, and I_x^i represents the intensity image of HR-SinSAR image patches. The numerical loss function is defined as follows:

$$L_{\text{val}}(\Theta) = \frac{1}{2N}\sum_{i=1}^{N}\left\|\Re^i - f_{\text{net}}\left(C_y^i, I_x^i\right)\right\|_{\text{F}}^2 \tag{9}$$

where Θ is the parameters of PSFN, $f_{\text{net}}(\cdot)$ is the output of the residual network, and $\|\cdot\|_{\text{F}}$ is the Frobenius norm.

Besides, we construct a polarimetric loss function from a physical imaging mechanism. For a given HR-PolSAR and HR-SinSAR image pair, the HR-SinSAR image can be regarded as the subset of the HR-PolSAR image in terms of polarimetric information. Therefore, for the PolSAR fusion results, the intensity image in a certain polarimetric mode should be consistent with the HR-SinSAR intensity image of the corresponding polarimetric mode. Therefore, considering the physical imaging relationship from the PolSAR images to the SinSAR images, the following polarimetric loss function can be used to constrain the network training process:

$$L_{\text{phy}}(\Theta) = \frac{1}{2N}\sum_{i=1}^{N}\left\|\left(I_x^i - \Im\left(C_u^i\right)\right) - \Im\left(f_{\text{net}}\left(C_y^i, I_x^i\right)\right)\right\|_{\text{F}}^2. \tag{10}$$

The total loss function of PSFN is expressed as follows:

$$L_{\text{total}} = \alpha L_{\text{val}}(\Theta) + \beta L_{\text{phy}}(\Theta). \tag{11}$$

Among them, the regularization parameters α and β are adaptively determined, and they satisfy the following relationship:

$$\alpha = \frac{L_{\text{val}}(\Theta)}{L_{\text{val}}(\Theta) + L_{\text{phy}}(\Theta)}, \beta = \frac{L_{\text{phy}}(\Theta)}{L_{\text{val}}(\Theta) + L_{\text{phy}}(\Theta)}. \tag{12}$$

The regularization parameters are adaptively adjusted through the numerical values of the loss functions, and higher weights are assigned to the larger loss so that the training process can converge faster.

Besides, according to the difference in polarimetric mode of input HR-SinSAR images, and under the condition of satisfying the numerical relationship of (3), the polarimetric loss can be decomposed into three independent subloss functions, as shown in the following:

$$L_{\text{HH}}(\Theta) = \frac{1}{2N}\sum_{i=1}^{N}\left\|\left(I_{\text{HH}}^i - \Im_{\text{HH}}\left(C_u^i\right)\right) - \Im_{\text{HH}}\left(f_{\text{net}}\left(C_y^i, I_{\text{HH}}^i\right)\right)\right\|_{\text{F}}^2 \tag{13}$$

$$L_{\text{HV}}(\Theta) = \frac{1}{2N}\sum_{i=1}^{N}\left\|\left(I_{\text{HV}}^i - \frac{\Im_{\text{HV}}\left(C_u^i\right)}{2}\right) - \frac{\Im_{\text{HV}}\left(f_{\text{net}}\left(C_y^i, I_{\text{HV}}^i\right)\right)}{2}\right\|_{\text{F}}^2 \tag{14}$$

$$L_{\text{VV}}(\Theta) = \frac{1}{2N}\sum_{i=1}^{N}\left\|\left(I_{\text{VV}}^i - \Im_{\text{VV}}\left(C_u^i\right)\right) - \Im_{\text{VV}}\left(f_{\text{net}}\left(C_y^i, I_{\text{VV}}^i\right)\right)\right\|_{\text{F}}^2 \tag{15}$$

where $L_{\text{HH}}(\cdot)$, $L_{\text{HV}}(\cdot)$, and $L_{\text{VV}}(\cdot)$, respectively, represent the polarimetric loss subfunctions when using I_{HH}, I_{HV}, and I_{VV} SinSAR data. $\Im_{\text{HH}}(\cdot)$, $\Im_{\text{HV}}(\cdot)$, and $\Im_{\text{VV}}(\cdot)$ represent the polarimetric intensity image extraction operators under polarimetric modes HH, HV, and VV, respectively. It should be noted that when the proposed fusion network uses specific polarimetric mode SinSAR data to assist in the reconstruction of LR-PolSAR images, only a single subpolarimetric loss function corresponding to the polarimetric mode is used.

III. Experimental Results

A. Experimental Dataset

Datasets used in the training process and test process are image pairs with four polarimetric channels PolSAR images and the auxiliary data of the SinSAR images. The data used come from two SAR imaging systems: RADARSAT-2 (RS-2) and AIRSAR. Due to the limitation of the amount of available data, the data used for network training are relatively limited. Therefore, this article only uses part of RS-2 images in San Francisco region for training, and the other is used to test. RS-2 images in Vancouver and Quebec region and AIRSAR images are used in the network test process. Due to the imaging characteristics of SAR systems, the spatial resolution of PolSAR images in azimuth and range direction is usually different. In addition, the difference in resolution may confuse the reconstruction effect of the two directions in visual evaluation. Therefore, in the experiment, we processed the azimuth and range to the same resolution by performing multilook processing on the SAR image. Besides, the radiometric calibration and despeckling are performed

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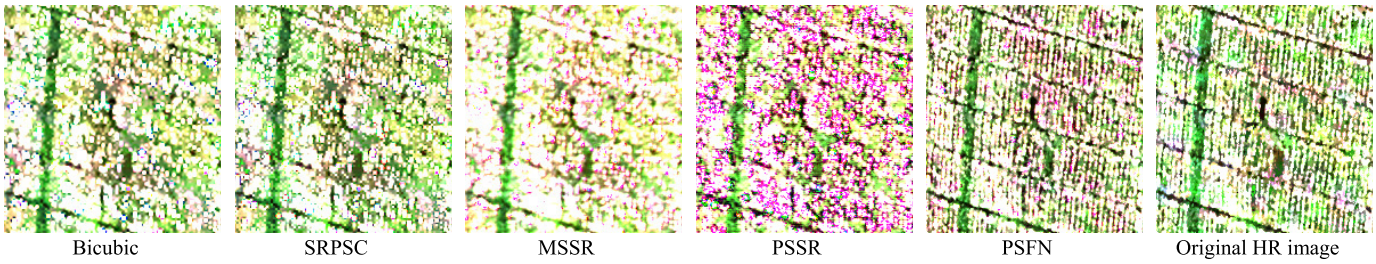

Fig. 10. Comparison results in the urban area.

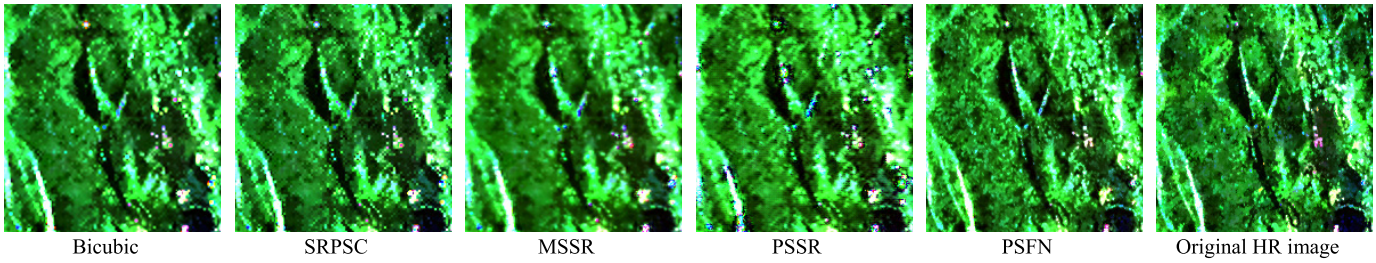

Fig. 11. Comparison results in the vegetation area.

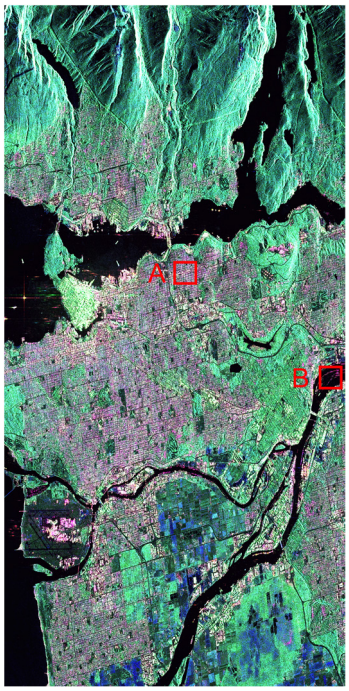

Fig. 12. PSFN fusion result for the RS-2 imagery (Vancouver).

as preprocessing by using PolSARpro_v5.0. Among them, the sigma calibration is used for radiometric calibration. We use nonlocal means and sigma Lee filter [36] to despeckling for LR-PolSAR images and HR-PolSAR images, while HR-SinSAR images are directly extracted from the despeckled HR-PolSAR images. The nearest-neighbor interpolation is used for down-sampling. The software parameters are the default settings.

1) Training Data: In terms of training data, two pairs of HR-PolSAR and HR-SinSAR image pairs are used to generate training data. The training data are all C-band data of RS-2. The train dataset details are listed in Table I. During the training process, the patch sizes of the PolSAR image and the SinSAR intensity image are $40 \times 40 \times 9$ and $40 \times 40 \times 1$, respectively. Besides, we used the PolSARpro V5.0 software provided by ESA to generate LR-PolSAR data with the patch size of $20 \times 20 \times 9$. In total, 71 040 patches were randomly generated and used for network training.

TABLE I
TRAIN DATASET INFORMATION

Sensor	Band	Region	Res	Size	mode
RS-2	C	San Francisco	8 m	6800×2400×9	HR PolSAR
			16m	3400×1200×9	LR PolSAR
			8 m	6800×2400×1	HR SinSAR
RS-2	C	San Francisco	8 m	5200×2400×9	HR PolSAR
			16m	2600×1200×9	LR PolSAR
			8 m	5200×2400×1	HR SinSAR

TABLE II
TEST DATASETS INFORMATION OF SIMULATED EXPERIMENTS

Sensor	Band	Region	Res	Size	mode
RS-2	C	San Francisco	16 m	1200×1200×9	LR PolSAR
			8 m	2400×2400×1	HR SinSAR
RS-2	C	Vancouver	16 m	3200×1600×9	LR PolSAR
			8 m	6400×3200×1	HR SinSAR
AIR SAR	L	San Francisco	24m	450×512×9	LR PolSAR
			12m	900×1024×1	HR SinSAR

2) Test Data: In terms of test data, three pairs of LR-PolSAR and HR-SinSAR image pairs are used for simulated experiments, which includes two scene image pairs of RS-2, and one scene image pair of AIRSAR. The standard mode SAR images and fine mode SAR images of RS-2 in Quebec region are used in real experiments.

The details of simulated experiments' test datasets are listed in Table II. The three datasets all use original HR-PolSAR images to down-sample and extract single-polarization channel intensity images to obtain LR-PolSAR images and high-resolution SinSAR images, respectively. In the simulation experiment, the input data of the three sets of data are LR-PolSAR and HR-SinSAR image pairs, and the acquisition conditions of the image pairs in each dataset are the same.

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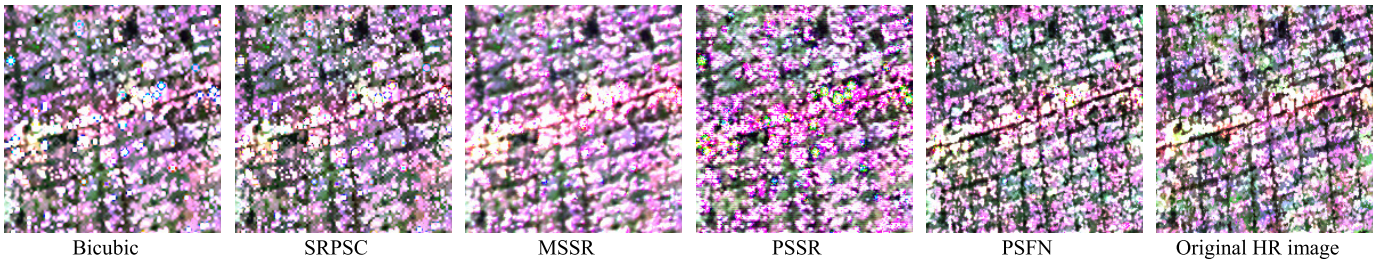

Fig. 13. Comparison results in the urban area.

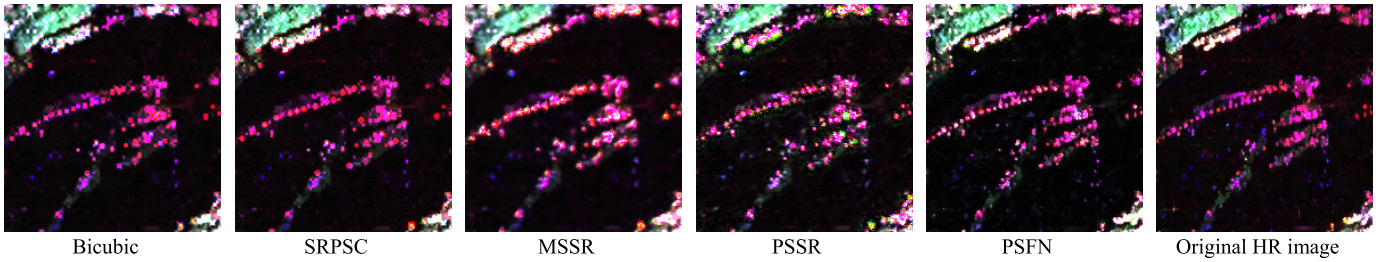

Fig. 14. Comparison results in the strong scattering point.

TABLE III
TEST DATASET INFORMATION OF REAL EXPERIMENTS

Satellite	RADARSAT-2	RADARSAT-2
Acquisition Type	Standard Quad-Pol	Fine Quad-Pol
Band	C	C
Looks	1	1
Antenna Pointing	Right	Right
Pass Direction	Descending	Ascending
Radar Center Frequency	5.41GHz	5.41GHz
Pulse Bandwidth	17.28MHz	30.00MHz
Incidence Angle Near Range	40.1°	40.1°
Incidence Angle Far Range	57.07°	67.82°
Nominal Spatial Resolution	25m	12m
Location	70.91-71.51E, 46.39-46.88N	71.06-71.51E, 46.61-46.88N

The details of the real experiment's dataset are listed in Table III. The dataset is obtained by RS-2 satellite in standard mode and fine mode, respectively. We performed radiometric calibration, despeckle processing, and the necessary registration processing on real experiments dataset. Besides, since the azimuth resolution of the images obtained in the fine mode and the standard mode is the same, to more intuitively demonstrate the effect of the proposed method, we have performed multilook processing and resample processing on the azimuth resolution of the standard mode data, making $r_{\text{azimuth}} = r_{\text{range}}$.

In the auxiliary data polarimetric mode selection, we compared the intensity images of RS-2 images in HH, VV, and HV polarimetric modes. As shown in Fig. 8, compared with the co-polarized channel, the intensity image of HV is only stronger in urban built-up areas, with lower values and weaker features in most areas. This phenomenon may make it difficult for us to extract features when extracting information from HV channel intensity images. Also, we noticed that the urban built-up area in the HH channel intensity image has a strong contrast with other areas. As auxiliary information, it may lead to excessive enhancement of urban built-up areas and weakening of other areas. Therefore, in the experiment of this article, we use the intensity image of the co-polarized channel VV with a more balanced intensity as input to ensure that the spatial information of the SinSAR image can be better utilized.

B. Parameter Settings

The proposed network includes four modules, LPSR, HSFE, HSSA, and LPCA. In Table IV, we list the detailed configuration information of each module of the network. We use Kaiming normal [37] to initialize the network model parameters. The Adam [38] is used to update weights. There are 200 epochs for network training. The initial learning rate is 0.0001, and it drops by half every 50 epochs. We use the PyTorch framework to train the fusion network in the Win10 environment, and an NVIDIA Quadro P4000 GPU. In order to meet the reproducibility requirements of the proposed method [39], we uploaded the source code of the network, and the link is (http://sendimage.whu.edu.cn/wp-content/uploads/2021/09/psfn-pytorch-master.zip).

C. Comparison Methods and Quantitative Evaluation Indices

We conducted quantitative evaluation experiments on various datasets to confirm the effectiveness of PSFN. Four PolSAR image reconstruction algorithms were selected for comparison, including Bicubic interpolation [40], SRPSC [14], MSSR [23], and PSSR [24]. Both MSSR and PSSR are retrained using the dataset used in this study. As there is no related work on the fusion of SinSAR and full-PolSAR, the comparison algorithms used are all super-resolution algorithms of full-PolSAR.

In the experimental analysis part, to intuitively reflect the physical mechanism corresponding to the research target in the PolSAR image rather than simply the degree of numerical retention, we carried out a data conversion on the reconstruction results of the PolSAR image, and transformed the C_3 matrix into a Pauli coherency matrix T_3 for subsequent quantitative evaluation experiments through the coherency matrix and covariance conversion equation [41], [42], as shown

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TABLE IV
CONFIGURATION OF PSFN

Module	Layer	Configuration
LPSR	Complex structure block	See the Complex structure block configuration below for details
	Transposed convolutional	TransConv (64×4×4, stride=2, pad=1) + PReLU
	ResBlock	See the ResBlock configuration below for details
HSFE	Convolutional	Conv1 (64×3×3, stride=1, pad=1) + PReLU:
	ResBlock	See the ResBlock configuration below for details
	Spatial attention block	See the Spatial attention block configuration below for details
	Convolutional	Conv2 (64×3×3, stride=1, pad=1) + PReLU:
HSSA	Convolutional	Conv1 (64×3×3, stride=1, pad=1) + PReLU
	Transposed convolutional	TransConv (64×4×4, stride=2, pad=1) + PReLU
	Convolutional	Conv2 (64×3×3, stride=1, pad=1) + PReLU + Sigmoid
	Convolutional	Conv3 (64×3×3, stride=1, pad=1) + PReLU:
LPCA	Convolutional	Conv1 (64×3×3, stride=1, pad=1) + PReLU
	Transposed convolutional	TransConv (64×4×4, stride=2, pad=1) + PReLU + Sigmoid
	Convolutional	Conv2 (64×3×3, stride=1, pad=1) + PReLU:
	Convolutional	Conv3 (64×3×3, stride=1, pad=1) + PReLU:
ResBlock	Convolutional	Conv1 (128×3×3, stride=1, pad=1) + PReLU
	Convolutional	Conv2 (64×3×3, stride=1, pad=1)
Spatial attention block	Convolutional	Conv1 (64×3×3, stride=1, pad=1) + PReLU
	Convolutional	Conv2 (1×3×3, stride=1, pad=1) + Sigmoid
	Convolutional	Conv3 (64×3×3, stride=1, pad=1) + PReLU

in (16) and (17). Subsequently, we perform Pauli decomposition of the coherency matrix to obtain three polarimetric components, as shown in (18). Then, calculate the quantitative index of the polarimetric component

$$T_3 = U_{3(L\to P)} C_3 U_{3(L\to P)}^{-1} \tag{16}$$

$$U_{3(L\to P)} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & \sqrt{2} & 0 \end{bmatrix} \tag{17}$$

$$P_1 = \frac{S_{\mathrm{HH}} + S_{\mathrm{VV}}}{\sqrt{2}},\ P_2 = \frac{S_{\mathrm{HH}} - S_{\mathrm{VV}}}{\sqrt{2}},\ P_3 = \frac{2S_{\mathrm{HV}}}{\sqrt{2}} \tag{18}$$

where P_1, P_2, P_3 represent the odd scattering (Odd) mechanism, the double scattering (Dbl) mechanism, and the volume scattering (Vol) mechanism, respectively.

In the experimental analysis, peak signal-to-noise ratio (PSNR) and mean absolute error (MAE) are employed to quantitatively assess the fusion results. Among them, the higher the PSNR value, the higher the quality of the images. The smaller the MAE, the more similar the predicted results are to the original high-resolution images. We calculated PSNR and MAE in polarimetric components of Pauli decomposition to evaluate spatial information reconstruction ability and polarimetric information retention ability of the algorithms.

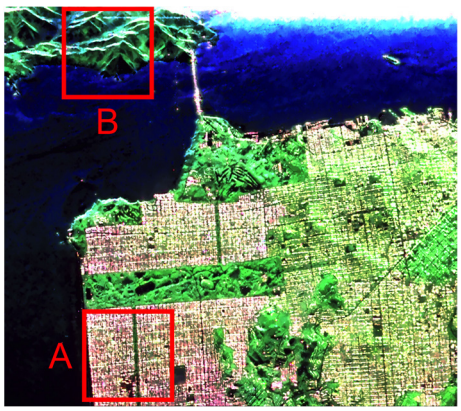

Fig. 15. PSFN fusion result for the AIRSAR imagery (San Francisco).

D. Simulated Experiments

1) Simulated Experiments of RS-2 in San Francisco Area: This set of experiments is used to verify the reconstruction effect of the fusion network under the same satellite and similar imaging environments, such as roughly the same imaging area and similar imaging angles. This set of experiments is mainly aimed at the application scenario of using the same sensor to monitor the same area for a long time series.

This set of experiments included a scene RS-2 image in San Francisco region which is displayed in Fig. 9. We selected two typical targets for visual evaluation, including urban areas and vegetation areas, as shown in Figs. 10 and 11. Compared with the deep learning-based methods, the results of bicubic interpolation are too smooth, while the results of the SRPSC method have an obvious grid effect. The results of the MSSR method have color distortion and are relatively smooth. The results of PSSR have artifacts in some areas. The proposed fusion network has a better ability to maintain texture information, and the color of the Pauli-composite images is similar to original resolution images. In the urban area, the proposed method keeps the shape of the buildings better. In the vegetation area, the results of PSFN are sharper at the edge and the ridgeline is also clearer. In Table V, the quantitative evaluation results illustrate that the proposed fusion network has a significant improvement in PSNR and MAE. It indicates that the results of the fusion network more resemble original high-resolution images in texture information and have a lower absolute error than the comparison algorithm.

2) Simulated Experiments of RS-2 in Vancouver Area: This set of experiments is mainly used to verify the effectiveness of the fusion network under an inconsistent imaging environment, which is aimed at the main application scenario of this article. That is, the fusion of PolSAR images acquired by a certain sensor under different imaging environments, such as the inconsistent imaging areas and inconsistent orbits.

This set of experiments included the scene RS-2 data in the Vancouver area which is displayed in Fig. 12. In this set of experiments, as shown in Figs. 13 and 14, urban densely built-up areas and strong scattering points were selected for

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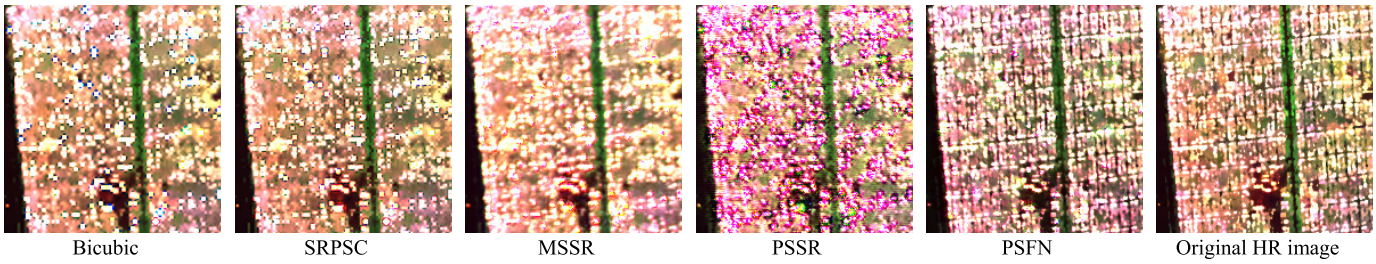

Fig. 16. Comparison results in the urban area.

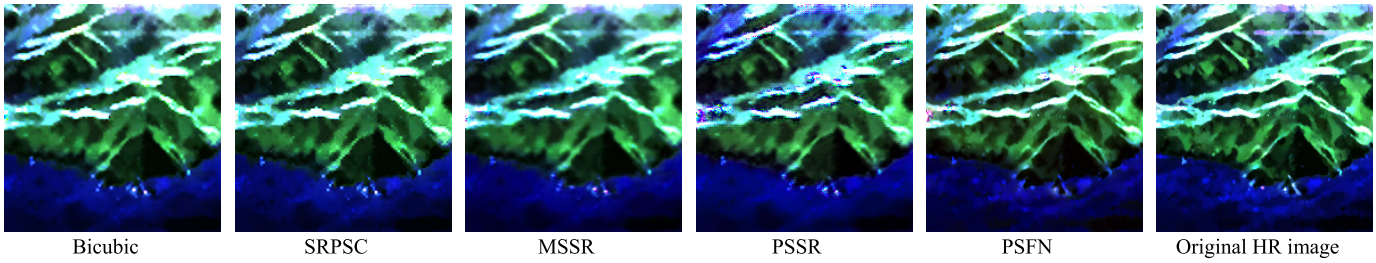

Fig. 17. Comparison results in the vegetation and sea area.

visual evaluation. Compared with the deep learning-based methods, the interpolation results are too smooth, and the results of the SRPSC method are mosaic-like due to the grid effect. The target boundary of the reconstruction results of the MSSR method is relatively smooth, and the PSSR method introduces obvious artificial artifacts while reconstructing the details. The proposed method can effectively reconstruct texture details, and the boundaries between targets are clear and sharp. In the urban densely built-up area, the proposed method can effectively reconstruct the vertical road, but all the comparison algorithms cannot reconstruct the detailed information of vertical roads well. In the area of strong scattering points, the comparison algorithms all have different degrees of defocusing. The adjacent strong scattering points overlap, and the blurred boundary makes it difficult to distinguish between the scattering points. The strong scattering points reconstructed by the proposed method have no obvious defocusing phenomenon. The sharpness of the boundary between adjacent strong scattering points is higher, and the strong scattering points can be identified easily, which is more conducive to image interpretation. In Table VI, the proposed fusion network has greatly improved in PSNR and MAE over comparison algorithms.

3) Simulated Experiments of AIRSAR in San Francisco Area: This set of experiments is mainly used to verify the fusion network generalization ability and reconstruction ability under different sensors and inconsistent imaging environments. It included the scene AIRSAR data, as displayed in Fig. 15. We selected urban areas, vegetation, and sea areas for visual evaluation.

As shown in Figs. 16 and 17, the color of the reconstruction results of the traditional methods is maintained well, but the spatial details are insufficient, which have an obvious grid effect. Compared with the traditional methods, the MSSR method eliminates the grid effect, but the spatial details are too smooth. The PSSR method has obvious artifacts. The proposed fusion network has a significant improvement in spatial

TABLE V

QUANTITATIVE EVALUATION RESULTS OF RS-2 (SAN FRANCISCO)

Method	Bicubic	SRPSC	MSSR	PSSR	PSFN
PSNR ($\|P_1\|^2$)	45.63	46.04	46.08	47.25	**52.04**
PSNR ($\|P_2\|^2$)	43.16	42.84	43.19	43.42	**48.82**
PSNR ($\|P_3\|^2$)	51.39	50.89	52.23	52.53	**53.26**
PSNR (mean)	46.73	46.59	47.17	47.73	**51.37**
MAE ($\|P_1\|^2$)	0.28	0.24	0.36	0.25	**0.08**
MAE ($\|P_2\|^2$)	0.20	0.19	0.25	0.21	**0.10**
MAE ($\|P_3\|^2$)	0.03	0.03	0.04	0.04	**0.03**
MAE (mean)	0.17	0.15	0.22	0.17	**0.07**

TABLE VI

QUANTITATIVE EVALUATION RESULTS OF RS-2 (VANCOUVER)

Method	Bicubic	SRPSC	MSSR	PSSR	PSFN
PSNR ($\|P_1\|^2$)	46.32	45.84	47.32	47.33	**53.08**
PSNR ($\|P_2\|^2$)	43.49	43.04	44.40	44.55	**51.08**
PSNR ($\|P_3\|^2$)	53.63	53.31	55.03	54.98	**56.49**
PSNR (mean)	47.81	47.40	48.92	48.95	**53.55**
MAE ($\|P_1\|^2$)	0.17	0.16	0.16	0.15	**0.07**
MAE ($\|P_2\|^2$)	0.15	0.14	0.16	0.13	**0.07**
MAE ($\|P_3\|^2$)	0.04	0.04	0.04	0.04	**0.03**
MAE (mean)	0.12	0.11	0.12	0.11	**0.06**

information, and texture information of the results is consistent with original resolution images. However, we noticed that the proposed method has color distortion in parts of urban areas, which may be caused by the large difference in the numerical dynamic range between different sensors and the lack of AIRSAR data during the training process. The quantitative evaluation is summarized in Table VII. The results of the PSFN have significantly improved in PSNR and MAE over comparison algorithms, which indicate the better generalization ability and stronger robustness of the proposed fusion network.

E. Verification of Real Experiments

To confirm the effect of PSFN in practical application, the fine mode SAR images and standard mode SAR images of RS-2 are adopted in real experiments.

In real experiments, we fused the LR-PolSAR images in the standard mode with the HR-SinSAR images in the fine mode,

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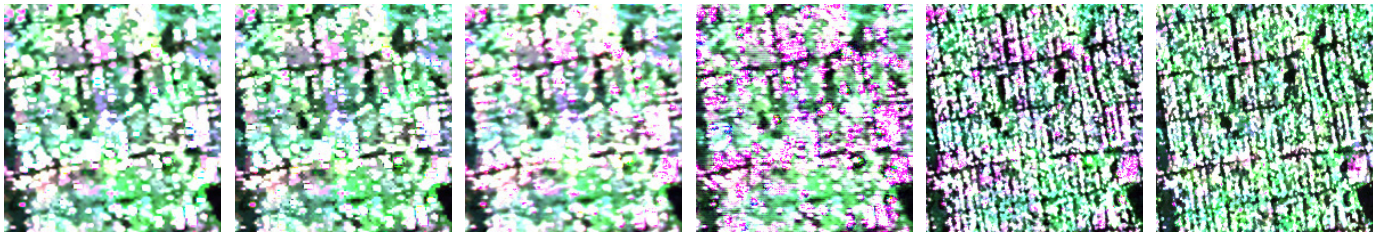
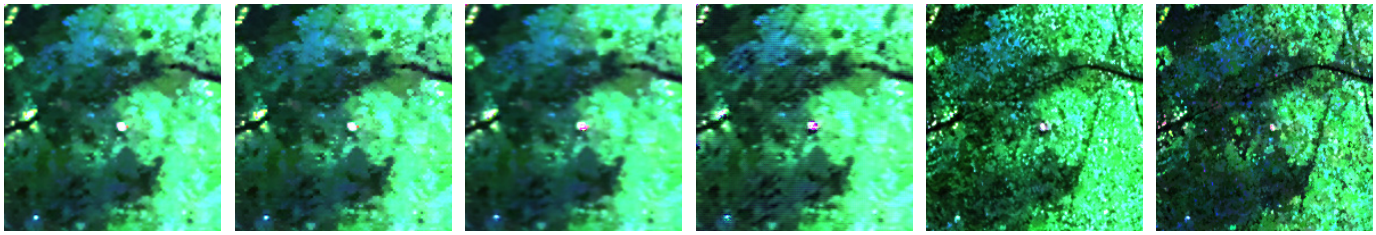
(a) (b) (c)

Fig. 18. PolSAR image in Quebec area. (a) Fine mode. (b) Standard mode. (c) PSFN.



Bicubic SRPSC MSSR PSSR PSFN Fine mode

Fig. 19. Comparison results in the densely built-up areas.



Bicubic SRPSC MSSR PSSR PSFN Fine mode

Fig. 20. Comparison results in the vegetation areas.

and calculated the quantitative indicator of the fusion results with the HR-PolSAR images in the fine mode. The PolSAR images were obtained in the fine mode and the standard mode, and the fusion results are listed in Fig. 18. Two typical land covers are selected for visual evaluation, including densely built-up areas and vegetation areas.

In Fig. 19, in densely built-up areas, the existing methods have poor detail information reconstruction, while the proposed method can better integrate the spatial detail information of HR-SinSAR images. In terms of color fidelity, the proposed method is more inclined to maintain the color of the data acquired in the standard mode. The reason is that in the network structure design, more consideration is given to maintaining the polarimetric information of the LR-PolSAR images rather than the polarization information of the single-polarization images. As shown in Fig. 20, in the vegetation areas, the existing methods are relatively vague, and the detailed information reconstruction is insufficient. The texture details of PSFN are more consistent with the PolSAR images in fine mode, and the color is more consistent with the PolSAR images in the standard mode. In terms of quantitative assessment, the proposed method improves average PSNR by more than 4.8 dB, and the average MAE decreases by more than 0.06 over existing methods, as shown in Table VIII. Visual evaluation and quantitative results illustrate that PSFN can effectively improve spatial resolution of LR-PolSAR images in practical applications.

In order to verify the effectiveness of the proposed method for small target reconstruction. We extracted two small targets from Quebec data for analysis, including islet and harbor. As shown in Fig. 21, in the experiment of the islet, the comparison algorithms showed different degrees of over-smoothing on the morphology of the islet, and the reconstruction of the proposed method is closer in morphology to the HR-PolSAR images obtained in the fine mode. As shown in Fig. 22, in the harbor experiment, the details of the reconstruction results of the comparison algorithms are less than those of the HR-PolSAR images, and the proposed method can reconstruct more detailed information.

In addition, we extracted the point spread functions (PSFs) of the reconstruction results of the small targets to verify the effectiveness of the proposed method. As shown in Figs. 23 and 24, the PSF extracted by the proposed method results is closer to the PSF extracted from the HR-PolSAR images under fine mode in terms of morphology and intensity than the comparison algorithm, indicating that the

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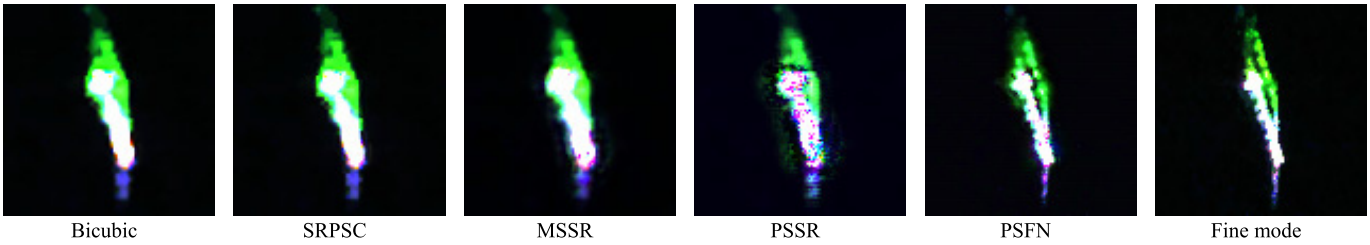

Fig. 21. Comparison results in the islet.

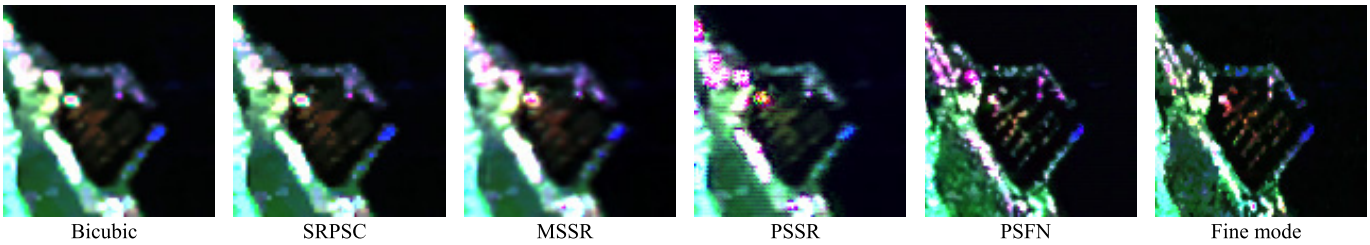

Fig. 22. Comparison results in the harbor.

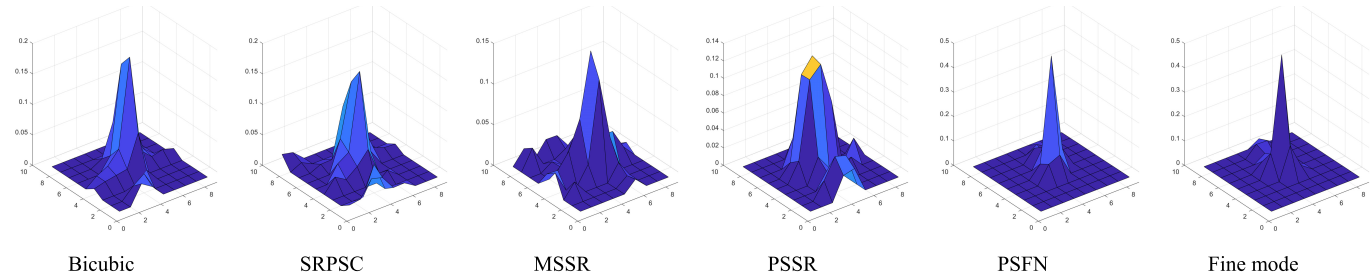

Fig. 23. PSF results in the islet.

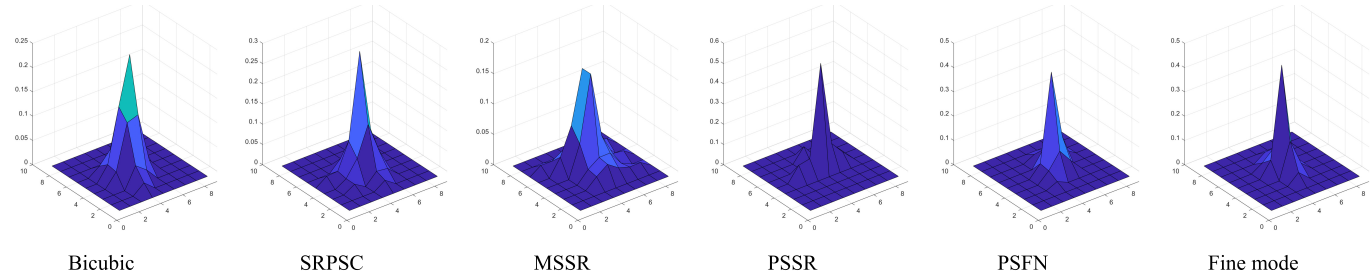

Fig. 24. PSF results in the harbor.

reconstruction effect of the proposed method is better than the comparison algorithms.

IV. Polarimetric Analysis

A. Analysis of Trace Moment ENL

To check whether the proposed method introduces artifacts in the fusion process, the equivalent number of looks (ENL) of PolSAR images is calculated by using the covariance or coherency matrix. The ENL is expressed as follows:

$$\mathrm{ENL} = \frac{\mathrm{Tr}(\langle T_3 \rangle)^2}{\langle \mathrm{Tr}(T_3 T_3) \rangle - \mathrm{Tr}(\langle T_3 \rangle \langle T_3 \rangle)} \tag{19}$$

where $\mathrm{Tr}(\cdot)$ is the trace operator. T_3 is the PolSAR coherency matrices, and $\langle \cdot \rangle$ is neighborhood averaging; in practical applications, it is generally recommended to use the 7×7 neighborhood to calculate ENL [43], [44]. The ENL is commonly adopted to check the speckle level of SAR images. A high ENL value indicates a better suppression of speckles, meaning that the homogeneous area is smoother. The PolSAR fusion results usually cannot reconstruct the detailed information that is completely consistent with the real HR-PolSAR images, and it is often smoother than the real HR-PolSAR images. Therefore, compared with original high-resolution images, the ENL value of fusion results is often higher. If the ENL of a certain area in the fusion results is lower than that of original high-resolution images, it indicates that the area has artificial artifacts.

We performed ENL calculations on the fusion results of RS-2 in the San Francisco region and mapped results. As shown in Fig. 25, the ENL map value of PSFN is higher, indicating that the results of PSFN are smoother than original high-resolution images, reflecting that no obvious artifacts are introduced in the reconstruction results of PSFN as a whole. To further verify the situation of artifacts, we selected densely built-up urban areas where artificial artifacts are easily introduced for ENL mapping and comparison. As shown

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TABLE VII
QUANTITATIVE EVALUATION RESULTS OF AIRSAR (SAN FRANCISCO)

Method	Bicubic	SRPSC	MSSR	PSSR	PSFN
PSNR ($\|P_1\|^2$)	50.29	51.44	47.72	53.93	**62.15**
PSNR ($\|P_2\|^2$)	44.94	45.62	46.08	46.49	**58.33**
PSNR ($\|P_3\|^2$)	51.09	51.58	51.49	52.00	**54.35**
PSNR (mean)	48.78	49.55	48.43	50.81	**58.28**
MAE ($\|P_1\|^2$)	0.16	0.09	0.36	0.15	**0.04**
MAE ($\|P_2\|^2$)	0.32	0.23	0.38	0.25	**0.04**
MAE ($\|P_3\|^2$)	0.11	0.08	0.18	0.12	**0.06**
MAE (mean)	0.20	0.13	0.31	0.18	**0.04**

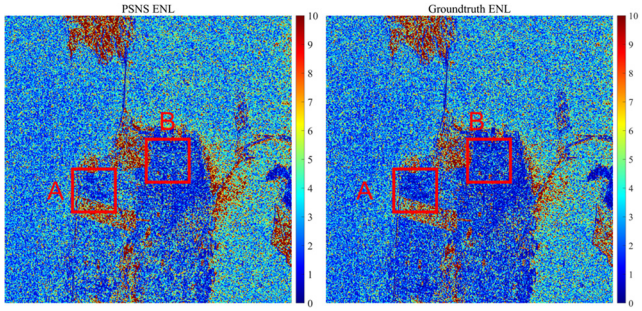

Fig. 25. ENL map results.

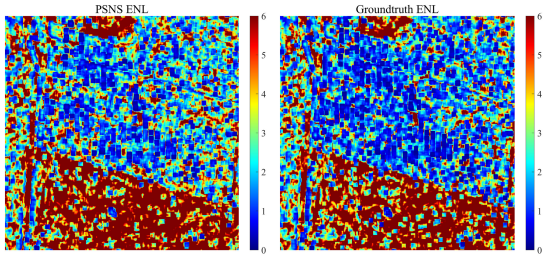

Fig. 26. ENL map results of area "A."

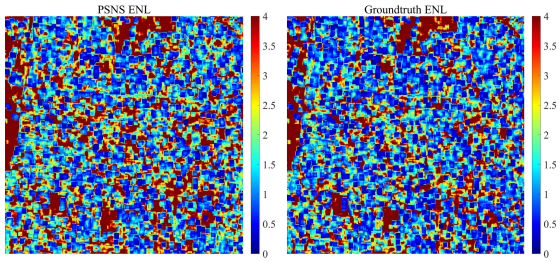

Fig. 27. ENL map results of area "B."

in Figs. 26 and 27, in the two selected densely built areas, the reconstruction results of the proposed method do not show that the ENL value is significantly lower than original high-resolution images, indicating that the number of artifacts in the reconstruction results is fewer.

B. Analysis of Polarimetric Decomposition

To confirm the polarimetric information retention ability, we performed polarimetric target decomposition on the reconstruction results to analyze the similarities and differences between the polarimetric decomposition results of the reconstructed image and the polarimetric decomposition results of original high-resolution images. Polarimetric decomposition is a theory that uses PolSAR polarimetric information to reveal the physical mechanism of scatters. By comparing the

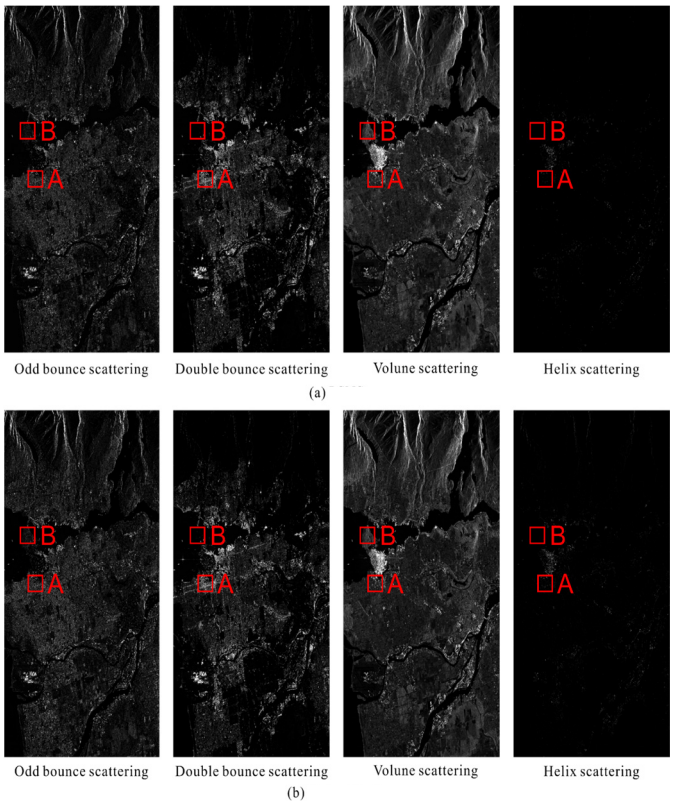

Fig. 28. Polarimetric decomposition results. (a) PSFN. (b) Ground truth.

scattering mechanism of fusion results with that of original high-resolution images under different scatters, it can reflect the degree of preservation of polarimetric information by the proposed method. In this article, the Yamaguchi decomposition (Y4R) [45] is used for polarimetric decomposition of fusion results, decomposed into Odd, Dbl, Vol, and helix scattering (Hlx).

In the RS-2 PolSAR image in the Vancouver area, as Fig. 28, we selected two typical types of land cover for evaluation, including densely built-up urban areas and vegetation areas. In the urban densely built-up areas, the wall of the building is perpendicular to the ground to form a dihedral angle. Therefore, Dbl is the dominant component of the scattering mechanism in this area. The scattering mechanism of densely built-up areas in urban is relatively complicated. As shown in Fig. 29, the echo signal has multiple scattering components in densely built-up areas, and single scattering and Vol also have a certain intensity in this area. Besides, compared with other areas, the intensity of Hlx in this area is higher. In the vegetation areas, under the action of the vegetation canopy, the scattering mechanism is mainly Vol. The intensity of single scattering is low, and there is almost no Dbl and Hlx. As displayed in Fig. 30, polarimetric decomposition results of the fusion results in the vegetation areas are consistent with that of corresponding original high-resolution images.

In order to intuitively show the performance of the comparison algorithms and the proposed method in maintaining polarimetric information, we used two quantitative indicators, correlation coefficient (CC) and MAE, to quantitatively evaluate the polarimetric decomposition results of the reconstructed

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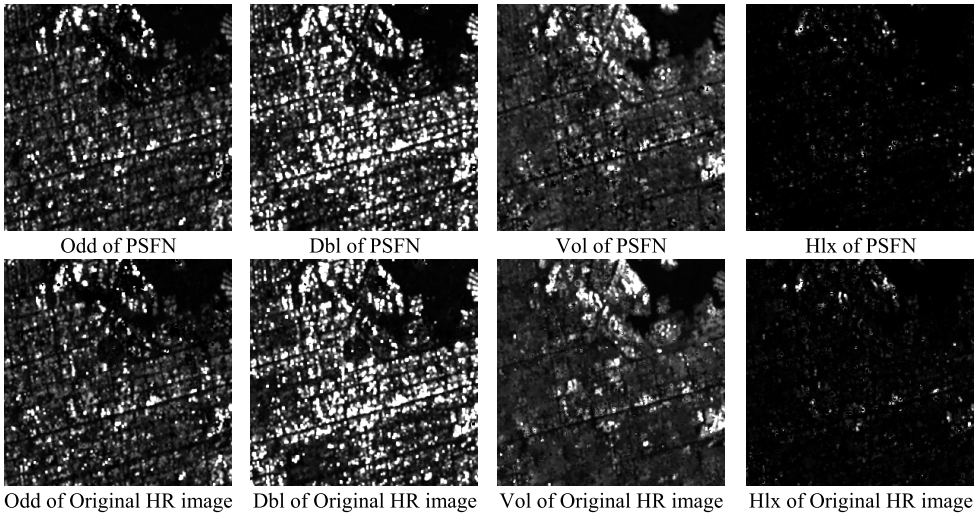


Fig. 29. Polarimetric decomposition results in the urban area.

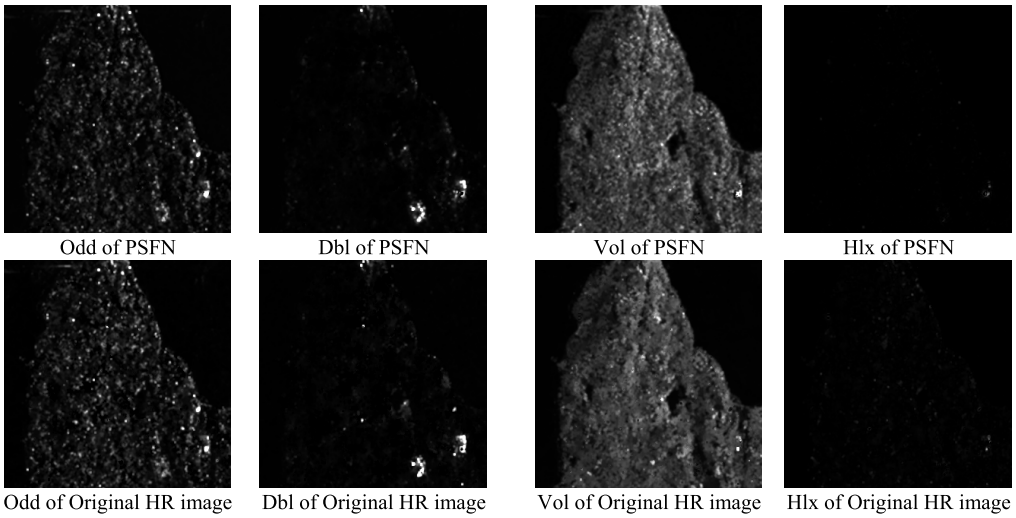
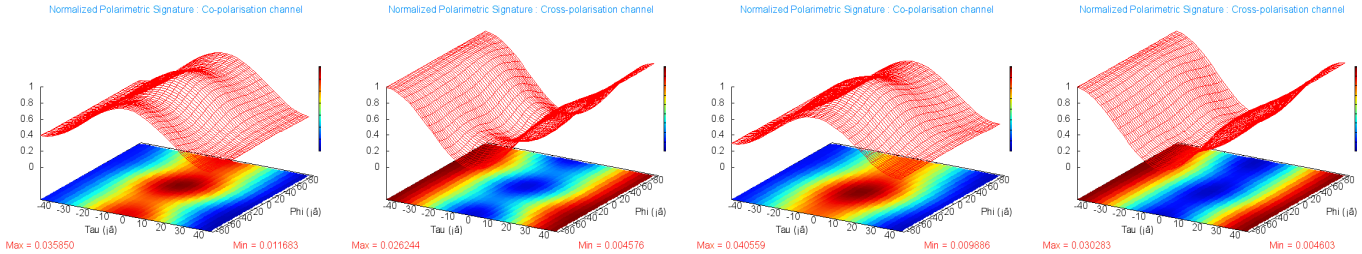


Fig. 30. Polarimetric decomposition results in the vegetation area.



(a) (b) (c) (d)

Fig. 31. Polarimetric signature in the vegetation area. (a) Co-polarimetric of PSFN. (b) Cross-polarimetric of PSFN. (c) Co-polarimetric of fine mode. (d) Cross-polarimetric of fine mode.

images and the original high-resolution images. It can be seen from Table IX that among the four polarimetric decomposition components, the CC and MAE of the Hlx components of all methods are relatively low. This phenomenon is because the Hlx component is not the main polarimetric scattering component, and the value of the helix component in most areas except urban areas is small or zero, so the CC and MAE of the helix component are relatively low. For the three main scattering components including Odd, Dbl, and Vol, the quantitative indicators of proposed method are superior to other algorithms. The proposed method has a higher correlation with the polarimetric decomposition results of the original high-resolution images. At the same time, the MAE of the proposed method is smaller, indicating that it can better maintain the polarimetric information.

C. Analysis of Polarimetric Signature

To analyze the extent to which the reconstruction results maintain the polarimetric characteristics, we selected two

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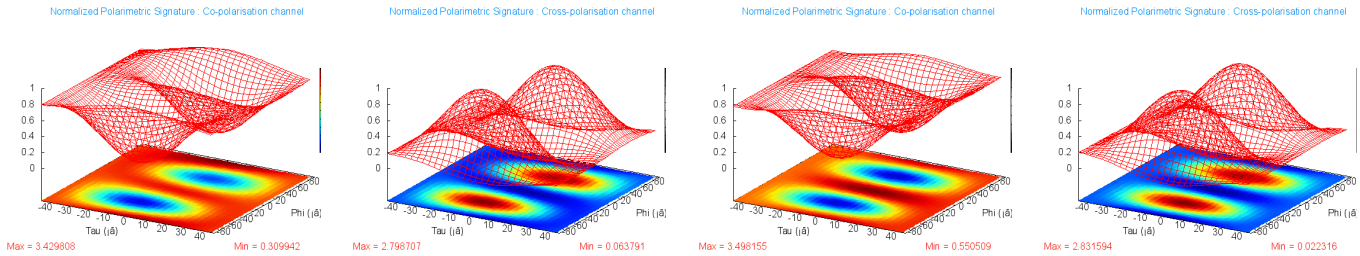

(a) (b) (c) (d)

Fig. 32. Polarimetric signature in the urban built-up area. (a) Co-polarimetric of PSFN. (b) Cross-polarimetric of PSFN. (c) Co-polarimetric of fine mode. (d) Cross-polarimetric of fine mode.

TABLE VIII
QUANTITATIVE EVALUATION RESULTS OF RS-2 (QUEBEC)

Method	Bicubic	SRPSC	MSSR	PSSR	PSFN
PSNR ($\|P_1\|^2$)	50.77	50.74	50.90	51.72	**56.29**
PSNR ($\|P_2\|^2$)	49.95	50.21	50.27	50.76	**58.39**
PSNR ($\|P_3\|^2$)	50.35	50.60	54.28	54.17	**56.59**
PSNR (mean)	50.36	50.52	51.82	52.22	**57.09**
MAE ($\|P_1\|^2$)	0.10	0.08	0.12	0.10	**0.03**
MAE ($\|P_2\|^2$)	0.10	0.09	0.14	0.11	**0.02**
MAE ($\|P_3\|^2$)	0.15	0.13	0.09	0.08	**0.05**
MAE (mean)	0.12	0.10	0.12	0.10	**0.04**

TABLE IX
QUANTITATIVE EVALUATION OF POLARIMETRIC DECOMPOSITION RESULTS

Method	Bicubic	SRPSC	MSSR	PSSR	PSFN
CC (Odd)	0.77	0.76	0.81	0.81	**0.92**
CC (Double)	0.83	0.82	0.86	0.85	**0.96**
CC (Volume)	0.82	0.81	0.84	0.80	**0.89**
CC (Helix)	**0.58**	0.58	0.53	0.36	0.56
MAE (Odd)	0.07	0.07	0.06	0.07	**0.04**
MAE (Double)	0.08	0.08	0.07	0.07	**0.05**
MAE (Volume)	0.05	0.05	0.05	0.06	**0.04**
MAE (Helix)	0.01	0.01	0.01	**0.01**	0.01

ground targets with different scattering characteristics from the reconstruction results of the RS-2 in the Quebec area and generated the polarimetric signature, including vegetation areas and urban built-up areas. Polarimetric signature is a curved surface describing the scattering characteristics of ground scatterers in any polarimetric state, which is composed of a co-polarimetric signature and a cross-polarimetric signature. To a certain extent, it reflects the change in the echo power of a target under a specific polarimetric combination.

Vol is the main scattering form in vegetation regions. For Vol, the maximum power of the co-polarimetric signature occurs in the case of linear polarimetric, which is when the ellipticity angle tau is 0. And its maximum power is not related to the polarimetric azimuth. For cross-polarimetric signature, the maximum power appears in the case of circular polarimetric (tau $= \pm 45°$), and the minimum power appears in the case of linear polarimetric. In Fig. 31, the polarimetric signature results of PSFN and HR-PolSAR images in fine mode are consistent, and they are in the Vol mode. For urban built-up regions, Dbl accounts for the main part. The minimum power value of the co-polarimetric signature appears in the linear polarimetric case (tau $= 0°$) and the polarimetric azimuth phi is $\pm 45°$. The cross-polarimetric signature graph shows the maximum power value at the corresponding position. As shown in Fig. 32, the polarimetric signature modes of PSFN and HR-PolSAR images in the fine mode in urban built-up areas all conform to the Dbl mechanism. Among the objects with different polarimetric scattering characteristics, the polarimetric signatures generated by the proposed method conform to the corresponding scattering mechanism. Besides, the polarimetric signature results of the proposed fusion network are close to the polarimetric signature results of the corresponding PolSAR image in fine mode.

V. CONCLUSION

We propose a PSFN for generating HR-PolSAR images. Different from the previous PSSR convolution neural network, we introduced HR-SinSAR images as auxiliary information to the fusion framework, which effectively improved the spatial quality of LR-PolSAR images. To further enhance the feature extraction capability of joint input data, we designed a CroAM to extract and cross-weight the dominant features of two input data so that the features of the input data can be fully extracted. Besides, the polarimetric loss functions based on the physical imaging mechanism are applied to neural network training, so that the proposed fusion network enhances the spatial texture details and achieves better color fidelity. The experiments on the RS-2 and AIRSAR datasets prove the effectiveness of PSFN. Compared with the existing PolSAR image resolution enhancement methods, the proposed fusion network can effectively use precise spatial information of SinSAR, and it can improve the PolSAR image quality significantly. Besides, the fusion network is easier to converge during training. Due to the limitation of data availability, the weakly supervised or unsupervised deep learning methods for adaptive learning of arbitrary sensors will be the future research trend. Besides, the existing neural networks are black-box models, and the interpretability of the model is poor. Introducing the physical mechanism of PolSAR images into the network is the key to improving network interpretability in the future.

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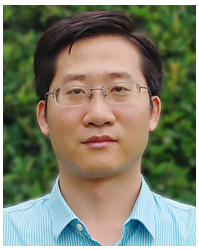
Liupeng Lin (Graduate Student Member, IEEE) received the B.S. degree in geographical information science from Wuhan University, Wuhan, China, in 2017, where he is currently pursuing the Ph.D. degree in cartography and geographic information engineering.

His research interests include image reconstruction, remote sensing image processing, and data fusion.

Jie Li (Member, IEEE) received the B.S. degree in sciences and techniques of remote sensing and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2011 and 2016, respectively.

He is currently an Associate Professor with the School of Geodesy and Geomatics, Wuhan University. His research interests include image quality improvement, image super-resolution reconstruction, data fusion, remote sensing image processing, sparse representation, and deep learning.



Huanfeng Shen (Senior Member, IEEE) received the B.S. degree in surveying and mapping engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

In 2007, he joined the School of Resource and Environmental Sciences (SRES), Wuhan University, where he is currently a Luojia Distinguished Professor and an Associate Dean of SRES. He was or is the PI of two projects supported by the National Key Research and Development Program of China and six projects supported by the National Natural Science Foundation of China. He has authored over 100 research articles in peer-reviewed international journals. His research interests include remote sensing image processing, multisource data fusion, and intelligent environmental sensing.

Dr. Shen is a Council Member of China Association of Remote Sensing Application, an Education Committee Member of Chinese Society for Geodesy Photogrammetry and Cartography, and a Theory Committee Member of Chinese Society for Geospatial Information Society. He is currently a member of the Editorial Board of *Journal of Applied Remote Sensing*, *Geography,* and *Geo-Information Science*.

Lingli Zhao (Member, IEEE) received the B.S. degree in geographic information system from Zhengzhou University, Zhengzhou, China, in 2010, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, Hubei, China, in 2015.

She was a Xiangjiang Scholar with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong, from 2017 to 2019. She is currently an Associate Professor with the School of Remote Sensing and Information Engineering, Wuhan University. Her research interests include polarimetric synthetic aperture radar (SAR) image processing and its applications in agriculture, disasters monitoring, and target interpretation and detection.

Qiangqiang Yuan (Member, IEEE) received the B.S. degree in surveying and mapping engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2006 and 2012, respectively.

In 2012, he joined the School of Geodesy and Geomatics, Wuhan University, where he is currently a Professor. He has authored more than 90 research articles, including more than 70 peer-reviewed articles in international journals such as the *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, IEEE TRANSACTIONS ON IMAGE PROCESSING, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include image reconstruction, remote sensing image processing and application, and data fusion.

Dr. Yuan was a recipient of the Youth Talent Support Program of China in 2019 and the Top-Ten Academic Star of Wuhan University in 2011. In 2014, he received the Hong Kong Scholar Award from the Society of Hong Kong Scholars and the China National Postdoctoral Council. He was a recipient of the recognition of best reviewers of the IEEE GRSL in 2019. He is an Associate Editor of five international journals, and has frequently served as a referee for more than 40 international journals for remote sensing and image processing.

Xinghua Li (Senior Member, IEEE) received the B.S. degree in geographical information system and the Ph.D. degree in cartography and geographical information engineering from Wuhan University, Wuhan, China, in 2011 and 2016, respectively.

In July 2016, he joined the School of Remote Sensing and Information Engineering, Wuhan University, as an Associate Professor. He is the PI of two National Natural Science Foundation of China (NSFC) projects and two subprojects of National Key Research and Development Program of China. He has authored more than 50 research articles. His research interests focus on multitemporal remote sensing analysis and application, remote sensing image processing, deep learning, and sparse representation.

Dr. Li is an Associate Editor of IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and *Journal of Imaging Science and Technology*, and a Guest Editor of *Journal of Applied Remote Sensing* and *Remote Sensing*. He has served as a reviewer for more than 70 SCI-indexed journals, such as *IEEE Geoscience and Remote Sensing Magazine* (GRSM), IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING (TGRS), IEEE TRANSACTIONS ON IMAGE PROCESSING (TIP), IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY (TCSVT), IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS), *Remote Sensing of Environment* (RSE), *International Journal of Remote Sensing* (IJRS), and *Journal of Applied Remote Sensing* (JARS).

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